

Hiver AI Customer Support Agent for AppleSupport

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GitHub Repository: https://github.com/Anuragpathak07/Hiver_Submission.git | **Decision Log:** refer to decision_log.md

Executive Summary & Headline Results

Evaluation Metric	System Result	Benchmark Baseline	Production Significance
Intent Classification Accuracy	85.00%	15.50% (Majority) 79.50% (TF-IDF)	+5.50% improvement over TF-IDF baseline across 8 intents.
Intent Classification Macro-F1	0.8512	0.0335 (Majority) 0.7954 (TF-IDF)	High balanced performance with zero class collapse.
Historical Retrieval Match @ 3	87.50%	69.50% (Lexical TF-IDF)	+18.00% improvement using semantic vector search.
Retrieval Top-1 Avg Similarity	0.7142	0.3711 (Lexical TF-IDF)	Accurately captures technical symptom paraphrasing.
LLM-as-Judge Quality Score	4.25 / 5.0	80.00% Pass Rate	Evaluated across 6 criteria (correctness, grounding, tone).
Human-Judge Agreement	90.00%	30-case validation	Strong alignment between automated judge & human review.
Auto-Handle Rate	73.00%	146 / 200 Golden Set	Safely resolves routine diagnostic & update queries.
Escalation Rate	27.00%	54 / 200 Golden Set	Safely escalates identity, billing, or low-confidence queries.

Core Engineering Principles & Production Philosophy

- **Comprehensive Multi-Layer Evaluation ('Proof Over System'):** Evaluated every tier independently—Intent Classifier, RAG Retrieval Engine, Grounded Generation, Deterministic Escalation, and LLM-as-Judge.
- **Production Safety & Risk-Aware Guardrails:** Our **27.0% Escalation Rate** reflects intentional safety design: high-risk queries (account security, billing) and low-confidence predictions (<0.50) route to specialists.
- **Engineering Integrity & Self-Awareness:** Includes a dedicated *'What is Misleading About My Headline Number?'* analysis dissecting confidence intervals (N=200), temporal drift (2017 iOS), and pre-annotation bias.
- **Instant Offline Reproducibility:** Evaluators can run unit/integration tests (pytest tests/) or single-query inference (python src/pipeline.py) in under 10 seconds without mandatory API keys.

1. Problem Framing & Brand Selection

1.1 Brand Selection Analysis (AppleSupport): Analyzed ~3 million tweets across brands. AppleSupport was chosen due to high multi-turn diagnostic volume (106,000+ tweets), technical symptom depth (iOS 11 battery drain, autocorrect glitch), and actionable resolution ground truth (support.apple.com URLs).

1.2 What 'Good' Means for AppleSupport: (1) Grounded Precision (no hallucinated steps), (2) Conservative Escalation (high-risk identity/billing queries routed to humans), (3) Reproducible Proof over trivial baselines.

2. Dataset Journey & Intent Discovery

2.1 Recursive Tree Reconstruction: Reconstructed flat tweets into **80,672 complete support cases** using parent tweet ID chains.

2.2 Bottom-Up Intent Discovery: Used all-MiniLM-L6-v2 dense embeddings + KMeans (\$k=12\$) + Groq LLM taxonomy consolidation to produce a frozen 8-intent taxonomy: *ios_update_issues*, *battery_life_issue*, *macbook_issues*, *keyboard_autocorrect_issue*, *music_audio_issue*, *iphone_hardware_issue*, *account_identity_issue*, *other_unclear*.

3. Hand-Annotation & Golden Set Methodology

Created a 200-example Golden Set benchmark via stratified sampling (seed=42). Maintained **strict data isolation** by excluding all Golden Set case IDs from training corpora and vector search indices. Human verification showed **92.50% agreement** with silver labels (15 disagreement edge cases corrected).

4. System Architecture

Customer Query → Intent Classifier (SentenceTransformer + LogReg 85% Acc)
→ Dual Vector Retrieval (87.5% Match@3)
→ Multi-Tiered Escalation Engine (Confidence < 0.50 or Sensitive Keywords → ESCALATE)
→ Grounded Response Generation (Groq LLM openai/gpt-oss-20b + Fallbacks)

5. Quantitative Results vs. Two Baselines

- **Majority Class Baseline:** Accuracy 15.50% | Macro-F1 0.0335
- **TF-IDF + Logistic Regression:** Accuracy 79.50% | Macro-F1 0.7954
- **SentenceTransformer + LogReg (Our Model):** Accuracy **85.00%** | Macro-F1 **0.8512** (+5.50% boost)

6. Top 5 Real Failure Modes Analysis

- **Hardware vs. Software Ambiguity:** Update-induced hardware symptoms (e.g. contacts glitch after iOS 11) misclassified as hardware.
- **Short / Noisy Customer Queries:** Expletive/short queries (e.g. 'what the fuck') yield low similarity (safely escalated).
- **Taxonomy Boundary Blur:** Store purchasing inquiries overlapping between account issues and general unclear queries.
- **Retrieval Response Divergence:** High query similarity where historical support response was an incomplete DM request.
- **Emoji & Informal Slang Over-conservatism:** High informal syntax variance lowering classifier confidence score.

7. 'What is Misleading About My Headline Number?'

- **Margin of Error on Small Golden Set (N=200):** 95% confidence interval spans [80.0%, 90.0%] (±5.0% error margin).

- **Stratified vs. Natural Class Distribution:** Stratified sampling overrepresents rare intents relative to real-world skewed Twitter traffic.
- **Silver Pre-Annotation Confirmation Bias:** Accepting silver pre-labels in 92.5% of cases introduces mild verification confirmation bias.
- **Temporal Shift (2017 Dataset):** Models trained on 2017 iOS 11 issues will degrade on modern iOS 17/18 queries without continuous retraining.
- **Retrieval Metric vs. Resolution:** Intent match @ 3 measures class overlap, not whether the historical response actually solved the user's issue.

8. Next-Week Improvement Plan

- **Blind Double-Annotation:** Annotate 1,000 cases blindly without silver pre-labels to eliminate confirmation bias.
- **URL Canonicalization:** Map raw 2017 t.co shortlinks to live support.apple.com canonical article endpoints.
- **Contrastive Fine-Tuning:** Fine-tune sentence-transformers using Multiple Negatives Ranking (MNR) loss on AppleSupport QA pairs.
- **LLM Response Filtering:** Filter out historical responses that are mere DM requests before populating vector index.